Catholic Junior College

JC 2 Preliminary Examinations
Higher 1

CHEMISTRY

8873/01

Paper 1 Multiple Choice

29 August 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, HT group and NRIC/FIN number on the Answer Sheet in the spaces provided.
Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

WORKED SOLUTIONS

For each question there are ***four*** possible answers, ***A***, ***B***, ***C*** and ***D***. Choose the **one** you consider to be correct.

1 Rubidium is a soft, metallic element that has a relative atomic mass of 85.47. There are two forms of naturally occurring rubidium isotopes, ^{85}Rb and ^{87}Rb.
^{87}Rb is slightly radioactive.
What percentage of naturally occurring rubidium is radioactive?

A 11.8 **B** 23.5 **C** 35.3 **D** 76.5

Concept: calculate the relative atomic mass of an element given the relative abundances of its isotopes

Answer: **B**

Let the % of ^{87}Rb be x.

Hence, the % of ^{85}Rb is $(100-x)$.

$$A_r \text{ of Rb} = \frac{87x+85(100-x)}{100} = 85.47$$

$x = 23.5\%$

2 10 cm^3 of hydrocarbon was burnt with 100 cm^3 of oxygen gas which is in excess. The resulting mixture was cooled to room temperature and the residual gases occupied a volume of 80 cm^3. After passing through sodium hydroxide, the volume of gas remaining is 40 cm^3.
What is the formula of the hydrocarbon?

A C_3H_6 **B** C_3H_8 **C** C_4H_8 **D** C_4H_{10}

Concept: perform calculations, including use of the mole concept, involving: volumes of gases (e.g. in the burning of hydrocarbons)

Answer: C

	C_xH_y +	$(x+y/4)\ O_2$ →	xCO_2 + $(y/2)\ H_2O$
Initial vol/cm^3:	10	100	0
Final vol/cm^3:	0	40	80-40=40
Change in vol/cm^3:	10	60	40
Ratio:	1	6	4

By inspection, x = 4,
(x+y/4) = 6,
Solving for y, y=8.
Therefore, the hydrocarbon is C_4H_8.

3 Which of the following is true when equimolar amounts of NH_3 and BF_3 is reacted?

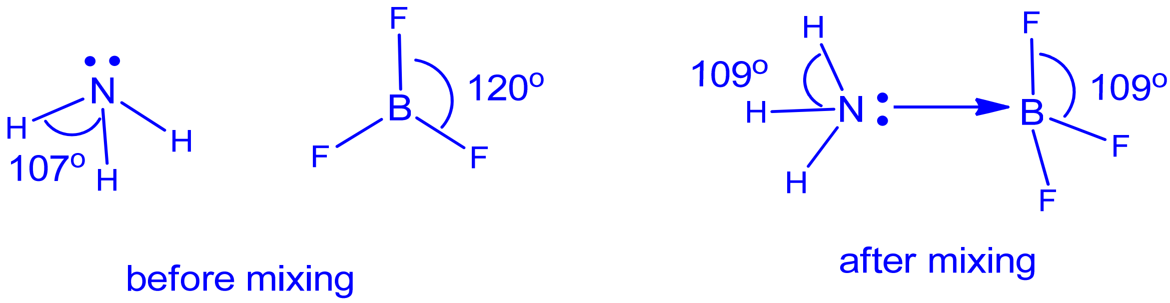
1 a dative covalent bond is formed between NH_3 and BF_3
2 N in NH_3 accepts a lone pair of electrons from B in BF_3
3 the shape around N in NH_3 changes from trigonal planar to tetrahedral
4 after mixing, the bond angle around B in BF_3 decreased from 120° to 109°

A 1 and 2 only
B 1 and 4 only
C 1, 3 and 4 only
D 3 and 4 only

Concept: dative covalent bonds, bond angles and shapes

Answer: B (1 and 4 only)

Since B in BF_3 only has 6 valence electrons, it can form a dative covalent bond with NH_3 where N donates a lone pair of electrons to B.



	NH_3	BF_3
Before mixing	Shape: trigonal pyramidal Bond angle: 107°	Shape: trigonal planar Bond angle: 120°
After mixing	Shape: tetrahedral Bond angle: 109°	Shape: tetrahedral Bond angle: 109°

4 *Use of Data Booklet is relevant to this question.*

How many hydrogen atoms are present in 111g of propanoic acid?

[*L* = Avogadro constant]

A *L* **B** 1.5*L* **C** 7.5*L* **D** 9*L*

Concept: MCS, understanding of Avogadro's constant

Answer: D

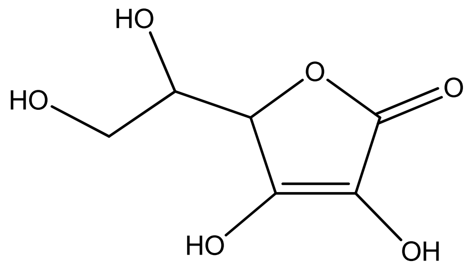
No. of moles of $CH_3CH_2CO_2H$ = 111 ÷ (12x3+ 6(1)+2(16)) = 1.5

$CH_3CH_2CO_2H \equiv 6H$

No. of moles of H = 6(1.5) = 9

No. of H atoms = 9*L*

5 Ascorbic acid, commonly known as vitamin C, is an important dietary supplement as it is essential in the repairing of tissues and can also function as an antioxidant.



vitamin C

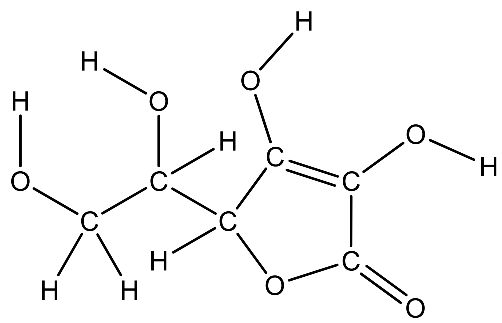
Which of the following shows the correct number of sigma and pi bonds present in one molecule of vitamin C?

	sigma bonds	pi bonds
A	11	2
B	16	2
C	20	1
D	20	2

Concept: identifying sigma and pi bonds in covalent bonds

Answer: D

All single bonds are sigma bonds. The double bonds are each one sigma and one pi. Hence, there are 20 sigma bonds and 2 pi bonds (1 in C=C and 1 in C=O) in one molecule of vitamin C.



6 The table below shows the boiling points of some halogenoalkanes.

halogenoalkane	boiling point/°C
CH_3CH_2Cl	12.3
CH_3CH_2Br	34.8
CH_3CH_2I	70.0

Which of the following statements correctly explains the difference in the boiling point?

A the bond energy of C-X bond decreases from C-C*l* to C-I

B the strength of instantaneous dipole-induced dipole attraction increases from CH_3CH_2Cl to CH_3CH_2I

C the strength of permanent dipole-permanent dipole attraction increases from C-C*l* to C-I

D the electronegativity difference between the halogen and carbon increases from C-C*l* to C-I

Concept: Chemical bonding (IMF), using IMF to explain for the trend of boiling point

Answer: B

A the bond energy of C-X bond decreases from C-C*l* to C-I
Statement is correct but boiling does not break the C-X bond, so this **does not** explain for the trend of **increasing** boiling point from CH_3CH_2Cl to CH_3CH_2I.

B the strength of instantaneous dipole-induced dipole attraction increases from CH_3CH_2Cl to CH_3CH_2I
Statement is **correct** as the total number of electrons increases from CH_3CH_2Cl to CH_3CH_2I and due to the increase in strength of id-id attraction. Hence, the boiling point increases from CH_3CH_2Cl to CH_3CH_2I.

C the strength of permanent dipole-permanent dipole attraction increases from C-C*l* to C-I
The C-C*l* bond is the most polar (compared to C-Br and C-I) hence the chloroalkane is expected to have the strongest intermolecular pdpd. Hence this statement is incorrect. But students need to note that in spite of the chloroalkane being expected to have the strongest intermolecular pdpd, the trend shows that the increase in idid strength due to the increase in number of electrons is the dominant factor that gives rise to the observed trend.

D the electronegativity difference between the halogen and carbon should **decrease** from C-C*l* to C-I
Also, the statement **does not** explain for the trend of increasing boiling point from CH_3CH_2Cl to CH_3CH_2I.

7 The following information is true for substance **Z**.

- It exists as a solid state at room temperature and pressure
- It has a regular lattice structure at room temperature and pressure
- It does not conduct electricity in both solid and molten state

Which of the following could be substance **Z**?

A iodine

B graphene

C aluminium

D sodium chloride

Concept: describe, interpret and/or predict the effect of different types of structure and bonding on the physical properties of substances

Answer: A

B graphene (conducts electricity in solid state)

C aluminium (conducts electricity in solid state)

D sodium chloride (conducts electricity in molten state)

8 *Use of the Data Booklet is relevant to this question.*

Radioactive carbon-14, ^{14}C, is often used to date archaeological and geological samples. Which species has the same number of neutrons and the same number of electrons as an atom of ^{14}C?

A $^{18}Ne^{2+}$ **B** $^{17}F^{+}$ **C** $^{16}O^{2+}$ **D** $^{14}N^{+}$

Concept: Atomic Structure, deduce the numbers of neutrons and electrons in given species.

Answer: C

	^{14}C	$^{18}Ne^{2+}$	$^{17}F^{+}$	$^{16}O^{2+}$	$^{14}N^{+}$
No. of Protons	6	10	9	8	7
No. of Neutrons	14 – 6 = 8	18 – 10 = 8	17 – 9 = 8	16 – 8 = 8	14 – 7 = 7
No. of Electrons	6	10 – 2 = 8	9 – 1 = 8	8 – 2 = 6	7 –1 = 6

9 *Use of the Data Booklet is relevant to this question.*

The successive ionisation energies, in kJ mol^{-1}, of an element **X** are given below.

940	2045	3060	4140	6590	7895	14990

What is **X**?

A $_{33}As$ **B** $_{34}Se$ **C** $_{39}Y$ **D** $_{85}At$

Concept: Atomic Structure, interpret successive ionisation energy data of an element in terms of the position of that element within the Periodic Table

Answer: B

The successive ionisation energies show a large increase from the sixth to seventh ionisation energy, hence there are six valence electrons, indicating that it should be element of Group 16.

10 Element **Y** is in period 3 and has melting point higher than the element before it. **Y** forms a chloride with low melting point which dissolves in water giving a solution of pH 2.
What is **Y**?

A magnesium **B** aluminium **C** silicon **D** phosphorus

Concept: Periodic Table
Answer: C

Magnesium, aluminium and silicon all have melting points higher than the respective elements before them. But only the chlorides of aluminium and silicon have low melting points (covalent chlorides). The pH of the solution of aluminium chloride will be about pH 4 or 3. Silicon chloride undergoes complete hydrolysis to produce HCl, hence resulting in a solution of pH 2 or less.

11 Which of the following statements about Group 17 elements and its compounds are true?

1 Atomic radius increases from chlorine to iodine mainly due to increasing number of electronic shells.

2 Iodine has a higher boiling point compared to chlorine because of stronger instantaneous dipole-induced dipole attraction

3 Hydrogen iodide is less stable to heat than hydrogen chloride due to weaker iodine-iodine bonds.

A 1 and 2 only
B 1 and 3 only
C 2 only
D 2 and 3 only

Concept: Periodic Table – Group 17
Answer: A

1 True. An atom of fluorine has 2 electron shells, chlorine has 3 electron shells etc.

2 Group 17 elements exist as non-polar molecules, which have intermolecular instantaneous dipole-induced dipole attractions. Due to the increase in number of electrons, the idid gets stronger down the group and more energy is required to overcome these. Hence, the higher boiling point of iodine compared to chlorine.

3 The thermal stability of hydrogen halides decreases down the group, due to decrease in bond strength of the H–X bond, not the X-X bond.

12 Which of the following will have the least exothermic lattice energy?

A barium oxide **B** magnesium oxide
C barium sulfide **D** magnesium sulfide

Concept: Energetics, effect of ionic radius on the numerical magnitude of lattice energy

Answer: C

LE $\propto -\left|\frac{q^+q^-}{r^++r^-}\right|$, in this case, both q^+ and q^- are the same at +2 and –2 respectively. However, comparing sulfide, S^{2-} and oxide O^{2-}, **sulfide will have the larger ionic radius**, r–, while barium ion, **Ba^{2+} will also have the larger ionic radius**, r^+, due to **increased shielding effect** as there are more electron shells which is **more significant than the increase in nuclear charge**, as compared to O^{2-} and Mg^{2+} respectively.

13 Hydrazine, N_2H_4, was used as fuel for the Apollo space mission, and it can be synthesised in an environmentally friendly manner according to the following reaction.

$$2NH_3(aq) + H_2O_2(l) \longrightarrow N_2H_4(l) + 2H_2O(l)$$

substance	$\Delta H_f^\ominus$/ kJ mol^{-1}
$H_2O(l)$	–286
$N_2H_4(l)$	+51
$NH_3(aq)$	–80
$H_2O_2(l)$	–174

Given the above enthalpy changes of formation, what is the enthalpy change of the above reaction in kJ mol^{-1}?

A −187 **B** −19 **C** +19 **D** +187

Concept: Energetics, apply Hess' Law to carry out calculations involving given simple energy cycles and relevant energy terms (restricted to enthalpy changes of formation

Answer: A

$\Delta H_r^\ominus = \sum\Delta H_f^\ominus(\text{products}) - \sum\Delta H_f^\ominus(\text{reactants})$
$= [+51 + 2(-286)] - [2(-80) + (-174)] = \underline{-187 \text{ kJ mol}^{-1}}$

14 *Use of the Data Booklet is relevant to this question.*

Methanol (M_r 32) is a preferred fuel for high performance racing cars. When 1.00 g of methanol was burnt to heat up a copper can with 200 g of water, it was found that the temperature of the water rose by 27 °C.

Assuming there is no heat lost to the copper can or surrounding, which value for the enthalpy change of combustion of methanol is given by these results?

A −8026 kJ mol^{-1}
B −722 kJ mol^{-1}
C −251 kJ mol^{-1}
D −113 kJ mol^{-1}

Concept: Energetics, calculate enthalpy changes with the use heat change = *mcΔT*
Answer: B

$$\Delta H_c = \frac{-mc\Delta T}{n} = \frac{-200\times4.18\times27}{\frac{1}{32}} = -722305 \text{ J mol}^{-1} = -722 \text{ kJ mol}^{-1}$$

15 Equal amounts of hydrogen and iodine were kept at a constant temperature of 723 K until equilibrium was reached. After which, rapid cooling was conducted to extract the iodine present for titration, and it was found that 0.07 mol of iodine was present.

$$I_2(g) + H_2(g) \rightleftharpoons 2HI(g) \qquad K_c = 48.2 \text{ at } 723 \text{ K}$$

What is the amount in moles of HI at equilibrium?

A 0.236 **B** 0.243 **C** 0.486 **D** 1.84

Concept: Equilibrium, calculation of the quantities present at equilibrium, given appropriate data.

Answer: C

$$K_c = \frac{[HI]^2}{[I_2][H_2]} = \frac{\left(\frac{n_{HI}}{V}\right)^2}{\left(\frac{n_{I_2}}{V}\right)\left(\frac{n_{H_2}}{V}\right)}$$

$= \frac{(n_{HI})^2}{(n_{I_2})(n_{H_2})} = \frac{(n_{HI})^2}{(0.07)(0.07)}$ = 48.2 (note that the volume need not be known in this case)

$\therefore n_{HI} = \sqrt[2]{48.2}$ = 0.486

16 Methanol can be manufactured industrially, using a copper oxide catalyst at temperature of 575 K, according to the following reaction.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \qquad \Delta H = -91 \text{ kJ mol}^{-1}$$

Which of the following statements are correct?

1 The catalyst increases the equilibrium constant.
2 The catalyst increases the equilibrium yield.
3 At lower temperature the equilibrium yield will increase.
4 At lower temperature the equilibrium constant will increase.

A 1 and 2 only
B 2 and 3 only
C 1 and 4 only
D 3 and 4 only

Concept: Equilibrium, effects of temperature, or the presence of a catalyst on an equilibrium and value of the equilibrium constant.

Answer: D
Catalyst increases the rate of reaction by increasing the rate constant, but does not affect equilibrium position or equilibrium constant.

Decrease in temperature in this case will result in the system to counteract the change by favoring the exothermic forward reaction, hence POE shifts to the right, and that change will also result in an increase in equilibrium constant (only change in temperature will change the equilibrium constant)

17 The Haber process is a key reaction in the industry for producing ammonia.
Which statement about Haber process is **not** correct?

A Iron is used as a heterogeneous catalyst to increase rate of production.
B Temperature of around 500 °C is used to increase the yield of ammonia.
C High pressure is preferred for this reaction as it increases the yield of ammonia.
D High pressure and temperature leads to an increase in the rate of production.

Concept: Equilibrium, describe and explain conditions used in the Haber process
Answer: B (not correct)
B Temperature of around 500 °C is used to **increase rate** and will **decrease the yield** of ammonia instead, as the forward reaction is exothermic in nature, which means that lower temperature will favour the forward reaction and results in higher yield instead.

18 What is the final pH of the solution when equal volumes of 0.030 mol dm^{-3} of $Ba(OH)_2(aq)$ is mixed with 0.030 mol dm^{-3} of $HCl(aq)$?

A 1.5 **B** 1.8 **C** 12.2 **D** 12.5

Concept: pH calculation

Answer: C

Let the volume of each solution used be *v* dm^3. Total volume after mixing = 2*v*

Amt of $Ba(OH)_2$ = Amt of HCl = concentration x volume = 0.030*v*

$Ba(OH)_2 \equiv 2OH^- \equiv 2HCl$

Amt of OH^-= 2 x 0.030v = 0.060v

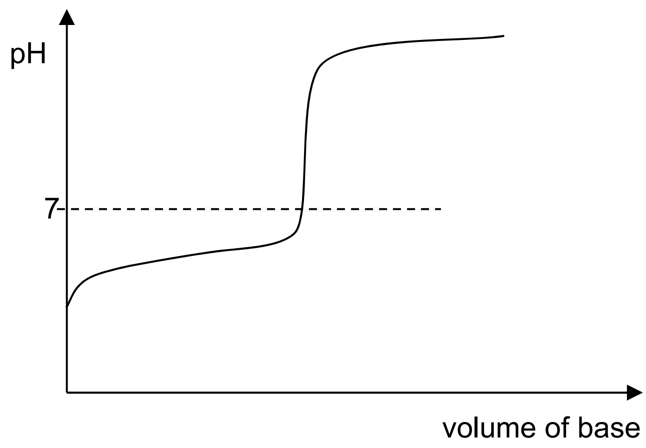
OH^- is in excess. Amt of OH^- that remains = $0.060v - 0.030v = 0.030v$

$[OH^-]$ in solution = $\frac{0.030v}{2v} = 0.015$ mol dm^{-3}

pOH = -lg(0.015) = 1.82

pH = 14 – 1.82 = 12.2

19 Visual indicators are often used to determine the end point of acid-base titrations. A titration was carried out between a weak acid and a strong base and the titration graph obtained is shown below:



Which of the following is a suitable indicator for the above titration?

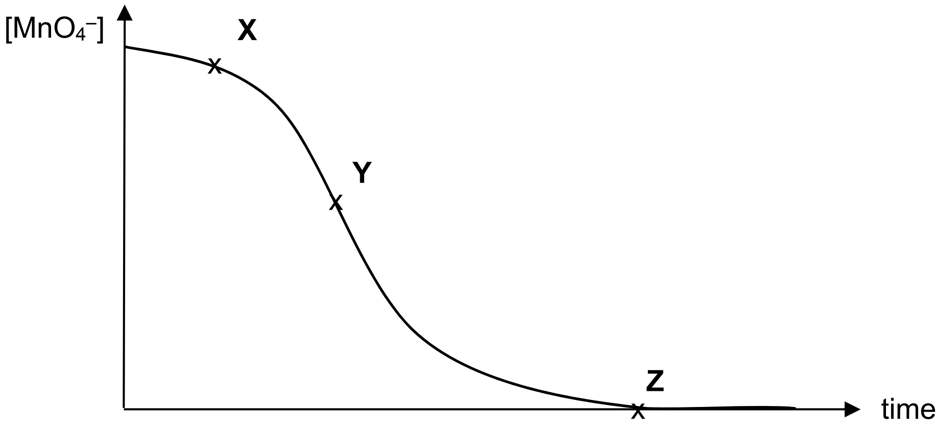
	Indicator	pH at which colour changes
A	Methyl violet	0 - 1
B	Bromocresol green	3 - 6
C	Bromothymol blue	6 - 7
D	Thymolphthalein	9 - 11

Concept: explain the choice of suitable indicators for acid-base titrations.
Answer: D
The indicator needs to have a distinct colour change were there is rapid pH change during the titration (vertical part of the graph). So a suitable indicator needs to change colour above pH 7.0.

20 The reaction of manganate(VII), MnO_4^-, with oxalate, $C_2O_4^{2-}$ is a type of autocatalytic reaction whereby the product, manganate(II) ions, is the catalyst for the reaction.

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

The following graph was obtained when $[MnO_4^-]$ was plotted against time.



Which of the following can be deduced from the graph?

A Rate at point **X** is very slow indicating that the temperature is low.

B Rate at point **Y** fast due to the production of the catalyst.

C Rate at point **Z** is slow as the catalyst is saturated.

D Rate at point **Z** is slow as the reaction is zero order with respect to $[MnO_4^-]$.

Concept: interpretation of concentration-time graph

Answer: B

Options A incorrect. Rate at point A is very slow as two anions are reacted together, not because the temperature is low.

Options C and D are incorrect because rate at point Z is zero as the reaction is complete.

21 When aqueous solutions of HCO_2H and Br_2 are reacted together, initial rate of reaction was found to be 0.080 mol dm^{-3} s^{-1}.
Given that the rate equation is:

$$\text{rate} = k[Br_2][HCO_2H]$$

Which of the following shows the initial rate of reaction when the original reaction mixture was mixed with an equal volume of water?

A 0.010 mol dm^{-3} s^{-1}
B 0.020 mol dm^{-3} s^{-1}
C 0.040 mol dm^{-3} s^{-1}
D 0.080 mol dm^{-3} s^{-1}

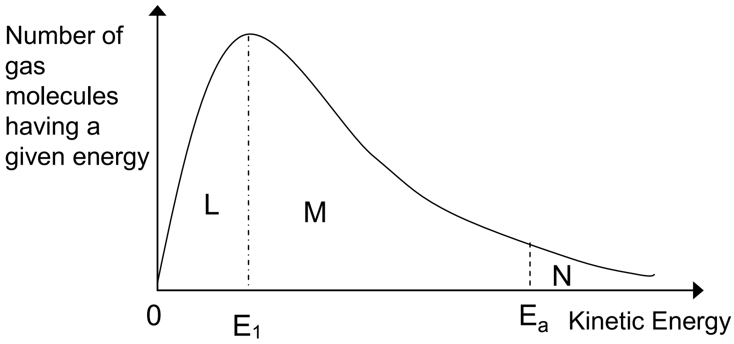
Concept: use rate equations to determine initial rate using concentration information
Answer: B
Let the original initial value of $[Br_2]$ be *x* and $[HCO_2H]$ be *y*.
When the reaction mixture was mixed with an equal volume of water, the concentration of each reagent falls to half the original value, i.e. ½x and ½y respectively.

New rate $= k \times \frac{1}{2}x \times \frac{1}{2}y$

$= \frac{1}{4}k\,xy$

$= \frac{1}{4}(0.080) =$ **<u>0.020 mol dm^{-3} s^{-1}</u>**

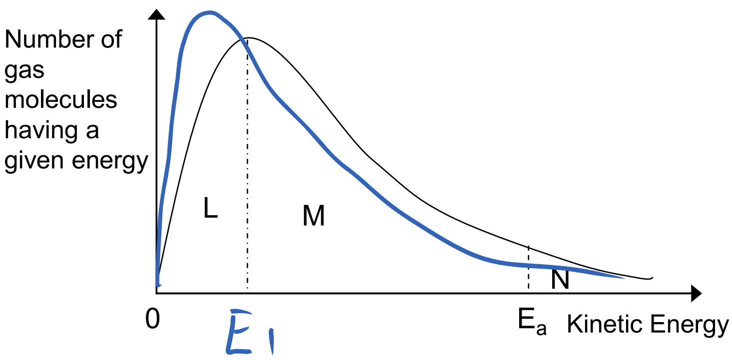
22 The Boltzmann distribution at a constant temperature is shown below:
E_1 and E_a are fixed energy values.



If the temperature is decreased by 10 ºC, what happens to the size of the areas labelled **L**, **M** and **N**?

	L	**M**	**N**
A	decreases	decreases	decreases
B	decreases	increases	decreases
C	increases	decreases	decreases
D	increases	decreases	increases

Concept: Explain in terms of the Boltzmann distribution the effect of temperature change.
Answer: C



23 The following table gives the approximate dimensions of certain substances.

	substance	**dimensions**
1	Fullerene (buckyball)	Diameter 0.71 nm
2	Red blood cell	Diameter of 8 μm, thickness of 2.5 μm
3	DNA strand	Diameter 2 nm, length up to 5 cm
4	Adenovirus	Diameter 90 nm

(1 μm = 1000 nm)

Which of these would be considered nanoparticles?

A 1 and 2 only
B 1, 2 and 4 only
C 1 and 4 only
D 2 and 3 only

Concept: Define the term nanoparticle
Answer: C
Nanoparticle – Material with all three dimensions on the nanoscale of 1-100 nm.
1 μm is 1000 nm, hence a red blood cell is already 8000 nm in diameter and 2500 nm in thickness.

24 Dishwasher pouches are small convenient packets that contain the detergent needed to be used in dishwashers. Such pouches are sturdy enough to be easily transported and safely handled by hand, yet dissolve easily when it is time to wash dishes. What could be the polymer that is used for the pouches?

A Polyvinyl alcohol
B Polyvinyl chloride
C Poly(diallyl phthalate)
D Poly(ethylene terephthalate)

Concept: Structure of polymer.
Answer: A
Only polyvinyl alcohol has the –OH functional group that can form extensive hydrogen bonding with water and is thus the most (only) water soluble polymer among the options.

25 The reaction scheme outlines the production of one of the monomers of nylon 66 from $Cl(CH_2)_4Br$.

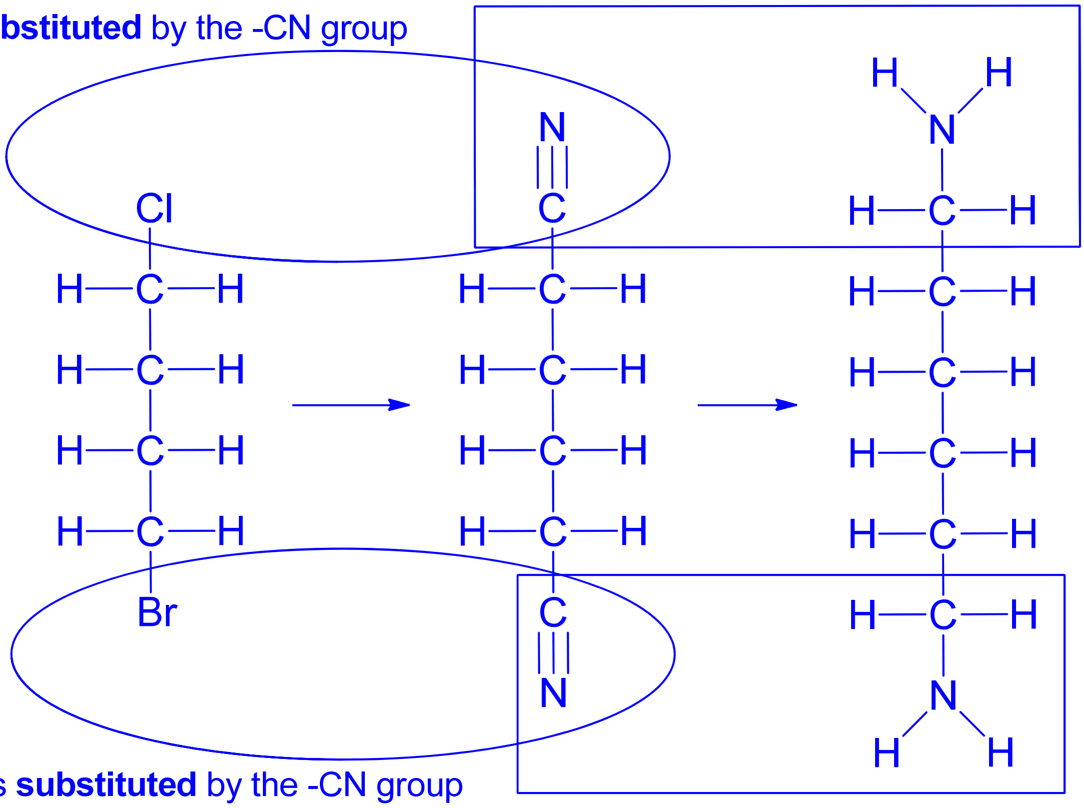
$$Cl(CH_2)_4Br \xrightarrow{\text{step 1}} NC(CH_2)_4CN \xrightarrow{\text{step 2}} H_2N(CH_2)_6NH_2$$

What are the types of reactions for each stage?

	step 1	step 2
A	substitution	oxidation
B	substitution	reduction
C	addition	addition
D	addition	reduction

Concept: interpret the type of reaction for organic reactions
Answer: B

-C*l* is **substituted** by the -CN group



-Br is **substituted** by the -CN group

Hydrogen added across the CN triple bond.
Hence **reduction** reaction.

26 Which of the following are true regarding polyester fabrics in comparison to polyamide fabrics?

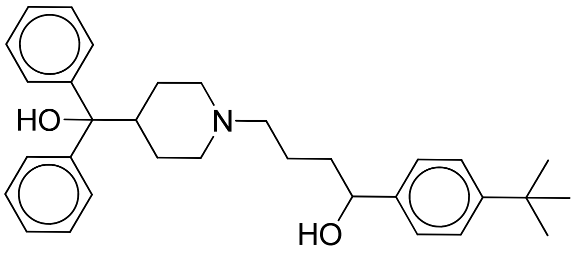
1. Polyester fabrics tend to crease less easily than polyamide fabrics.
2. Polyester fabrics tend to be more easily stained by oil than polyamide fabrics.
3. Polyester fabrics tend to absorb more sweat than polyamide fabrics.

A 1 and 2 only
B 2 and 3 only
C 3 only
D 1, 2 and 3 only

Concept: Polyamide and polyester structures in terms of interaction with water.
Answer: A
Polyesters have the ester functional group that is able to form less extensive hydrogen bonding compared to the amide functional group.

Refer to page 11 of Polymer part 3 notes:
Hence, polyesters do not absorb water well (statement 3 is thus incorrect and statement 2 is correct) and hence are used to produce fabrics to make clothes that are generally dry faster and are wrinkle free (statement 1 is correct).
Refer to page 13 of Polymer part 3 notes:
As polyamide chains are bonded to one another via hydrogen bonding, washing polyamide fabrics allow water to form hydrogen bonds with the polymer chains and temporarily change the way the polymer chains are lined up. As the water evaporates, new hydrogen bonds lock in places any creases that formed when the fabric was wet.

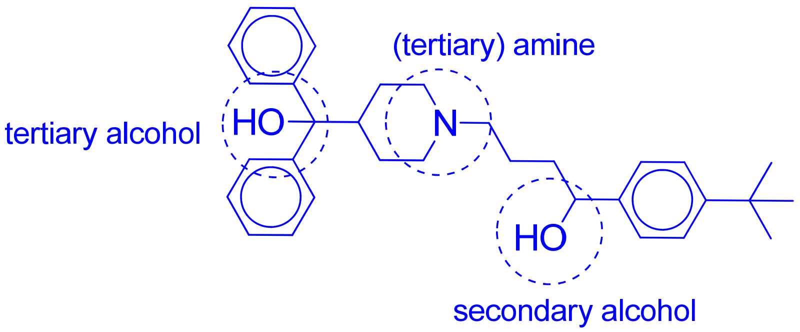
27 *Terfenadine* is an antihistamine formerly used for the treatment of allergic conditions. The structure of *terfenadine* is shown below:



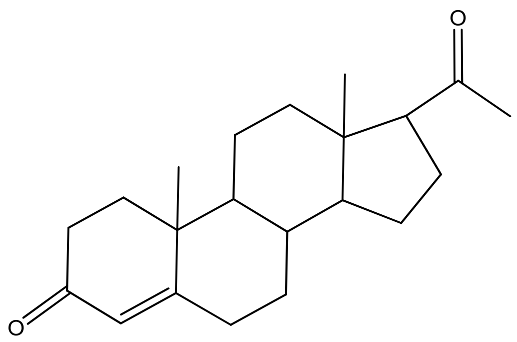
Which functional group is not present in *terfenadine*?

A amine
B tertiary alcohol
C secondary alcohol
D phenol

Concept: Interpret the structural formula of organic classes of compounds..
Answer: D



28 Progesterone is a hormone which is involved in menstrual cycle and pregnancy. The following structure shows one molecule of progesterone.



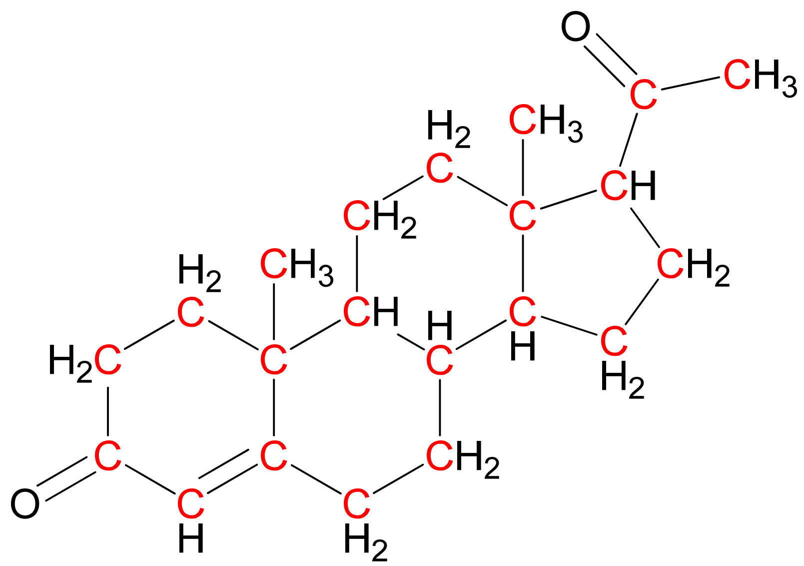
What is the molecular formula for the progesterone hormone?

A $C_{19}H_{14}O_2$ **B** $C_{19}H_{22}O_2$ **C** $C_{20}H_{30}O_2$ **D** $C_{21}H_{30}O_2$

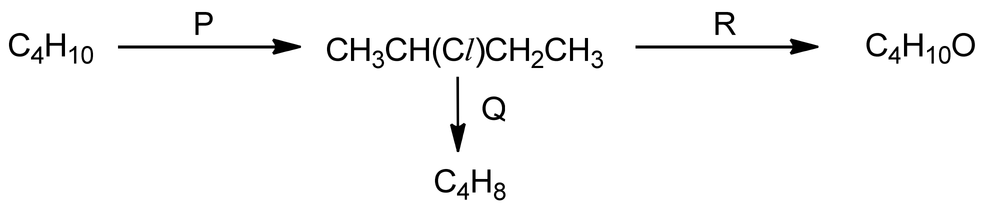
Concept: Skeletal formula, Mr of organic compounds

Answer: D

There should be 21 carbons (in red) and 30 H.



29 The diagram shows a reaction scheme involving 2-chlorobutane, $CH_3CH(Cl)CH_2CH_3$.



Which of the following is true regarding the reaction scheme above?

1 both reactions P and R are substitution reactions
2 reaction Q is a reduction reaction
3 NaOH can be used in both reactions Q and R
4 two constitutional isomers with the molecular formula of C_4H_8 can be formed from reaction Q

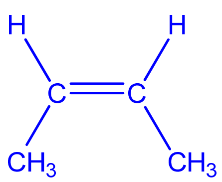
A 1 and 2 only
B 1 and 3 only
C 1 ,3 and 4 only
D 2, 3 and 4 only

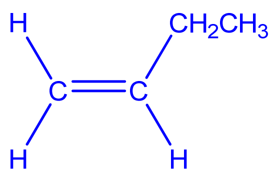
Concept: reaction of alkane and halogenoalkanes

Answer: C (1,3 and 4 only)

	Reaction P	**Reaction Q**	**Reaction R**
Type of reaction	**substitution**	**elimination**	**substitution**
Reagent and condition	**Cl_2 under uv light**	**NaOH(alcoholic), heat under reflux**	**NaOH(aq), heat under reflux**

reaction Q can produce two constitutional isomers:





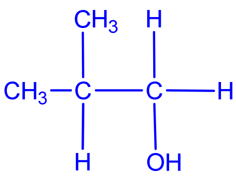
30 Which of the following alcohols can be formed when a ketone is reduced with $NaBH_4$ dissolved in methanol?

A 2-methylpropan-1-ol
B 2-methylbutan-2-ol
C 3-methylbutan-2-ol
D 2,3-dimethylbutan-2-ol

Concept: nomenclature of alcohols, classification of primary, secondary and tertiary alcohols, reduction of ketones to form secondary alcohols

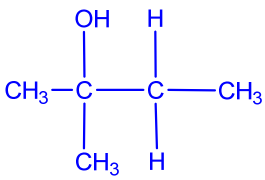
Answer: C

Ketones will be reduced to only **secondary** alcohols.



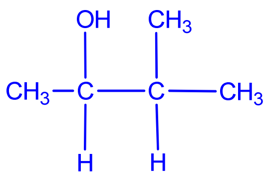
2-methylpropan-1-ol

primary alcohol



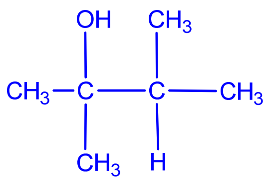
2-methylbutan-2-ol

tertiary alcohol



3-methylbutan-2-ol

secondary alcohol



2,3-dimethylbutan-2-ol

tertiary alcohol